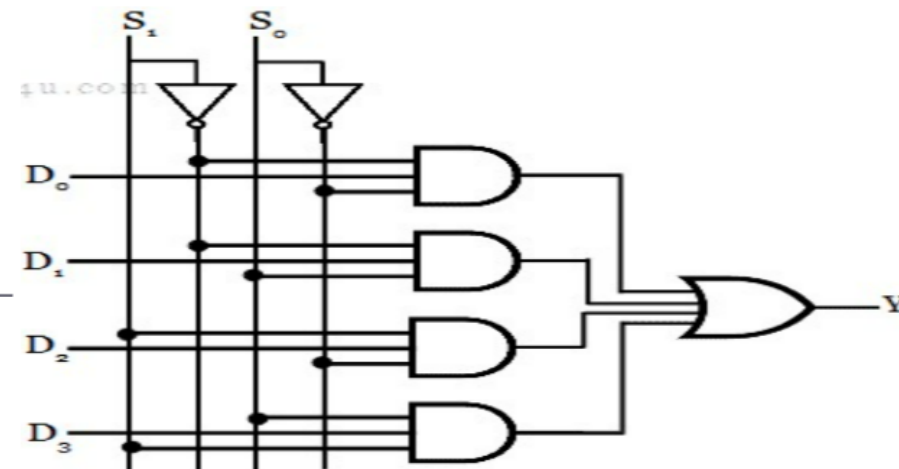


D_3	D_2	D_1	D_0	S_1	S_0	Y
1	1	1	0	0	0	0
1	1	0	1	0	0	0
1	0	1	1	0	0	0
0	1	1	1	0	0	0
1	1	1	1	0	1	1
1	1	1	1	1	0	1
1	1	1	1	1	1	1

```
//Gate_level
module mux4_gate(
    input D0, D1, D2, D3, s0, s1,
    output Y
);

    assign Y = ( D0 & (~s1) & (~s0) ) |
               ( D1 & (~s1) & ( s0) ) |
               ( D2 & ( s1) & (~s0) ) |
               ( D3 & ( s1) & ( s0) );

endmodule
```



```
//Behavioral
module mux4_behav(
    input D0, D1, D2, D3, s0, s1,
    output reg Y
);

    always @(*)
    begin
        case({s1,s0})
            0: Y= D0;
            1: Y= D1;
            2: Y= D2;
            3: Y= D3;
        endcase
    end

endmodule
```